

POS 401 Final Project

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Date 12/04/2025

Research Question:

How did the COVID-19 pandemic and Japanese Yen exchange rates impact tourism in Japan?

Literature Review

Traveling abroad is part of many people's goals, and numerous countries including Japan, use tourism as part of their economic growth and global status. However, the COVID-19 pandemic has impacted tourism, fluctuating cost and availability in many countries. Focusing on Japan, it was only after the pandemic in 2022, that Japan started changing regulations on tourism control, a long awaited and necessary change. According to Voice of America News in 2022, Japanese former prime minister Fumio Kishida stated that Japan will abolish many regulations regarding the COVID-19 pandemic, including border control, removal of pre-arrival PCR tests, and on the number of tourists, as long as they are negative, and vaccinated three times prior to the trip. (VOA news, 2022) It is stated that the Japanese Yen has hit the lowest mark in a quarter century, and these actions would promote the Japanese economy.

Hypothesis

- Null Hypothesis(H0): There is no statistical significance between COVID periods, Yen exchange rates, and the number of visitors in Japan.
- Alternative Hypothesis(H1): There is statistical significance between COVID periods and the number of visitors in Japan.
- Alternative Hypothesis(H2): There is statistical significance between Yen exchange rates and the number of visitors in Japan.

Methodology

In this short project, we will be taking a closer look at the tourism in Japan, by performing a regression analysis using two different datasets. The first dataset is the number of foreign visitors by month, and the second dataset is the Japanese currency exchange rate to USD dataset. As the research question has stated, one of our objectives is to see how the COVID-19 pandemic has impacted the number of tourists visiting Japan. In order to do so, we will be performing a simple linear regression to see the number of visitors in terms of three different periods, pre, mid, and post pandemic. The second model will also be using a linear model, comparing the exchange rate of Yen to Dollar, to the number of visitors in Japan. We want to find out if the fluctuation of Yen impacted tourism, and if so by how much compared to the COVID period variable.

Results 1

The first independent variable is COVID, more specifically the timeline of the pandemic. Therefore we split the variable into three main time periods, pre, mid, and post.

```
> head(foreign)
```

	Country.Area	Month	Year	Visitor
1	Afghanistan	Jan.	2017	233
2	Afghanistan	Feb.	2017	196
3	Afghanistan	Mar.	2017	192
4	Afghanistan	Apr.	2017	259
5	Afghanistan	May	2017	228
6	Afghanistan	Jun.	2017	123

```
> head(foreign)
```

	Country.Area	Month	Year	Visitor	Month_num	Date	COVID_period
1	Afghanistan	Jan.	2017	233	1	2017-01-01	Pre
2	Afghanistan	Feb.	2017	196	2	2017-02-01	Pre
3	Afghanistan	Mar.	2017	192	3	2017-03-01	Pre
4	Afghanistan	Apr.	2017	259	4	2017-04-01	Pre
5	Afghanistan	May	2017	228	5	2017-05-01	Pre
6	Afghanistan	Jun.	2017	123	6	2017-06-01	Pre

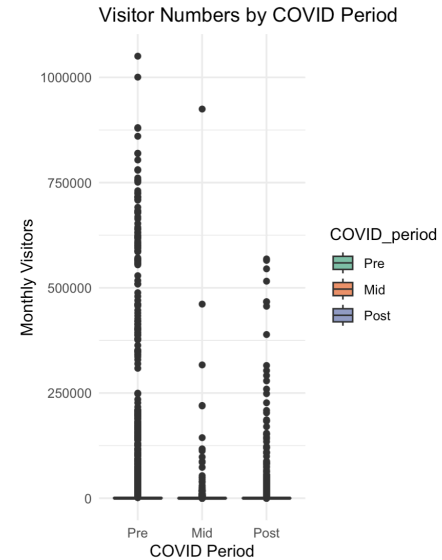
Because the format of the month and year were complicated to classify into periods, I decided to format them in “20xx-0x-xx”. Additionally, because our independent variable is covid period, I classified all of the visitors before 2020-01-01 as “Pre”, after that date until 2022-01-01 as “Mid” and any numbers after that period as “Post”.

```
Call:
lm(formula = Visitor ~ COVID_period, data = merged)

Residuals:
    Min       1Q   Median       3Q      Max
-12557  -12421  -2983   -894 1037863

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    12556.8     583.4   21.52  <2e-16 ***
COVID_periodMid -11661.5     922.4  -12.64  <2e-16 ***
COVID_periodPost -9571.5     922.4  -10.38  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 49870 on 17049 degrees of freedom
Multiple R-squared:  0.01122, Adjusted R-squared:  0.01111
F-statistic: 96.76 on 2 and 17049 DF, p-value: < 2.2e-16
```



After running the linear model, the summary showed that the “Pre” period had the most number of visitors, the estimate showed 12556.8. Relatively to the “Pre”, “Mid” period had a substantial drop in the number of visitors showing -11661.5, presumably so due to the border regulations and policy changes during the pandemic. As mentioned in the literature review, post 2022 was when the Japanese government started restarting their policies and open back up. Accordingly, the “Post” period showed -9571.5, which is still relative to the intercept, which is “Pre”. We can conclude that the number of visitors improved compared to “Mid” but didn’t reach the pre-pandemic numbers. All p-values showed statistical significance, although the R-squared showed around 0.01. It is understandable, as the limitations of this simple model cannot show all the outliers and variables of causing the decrease of tourism.

Results 2

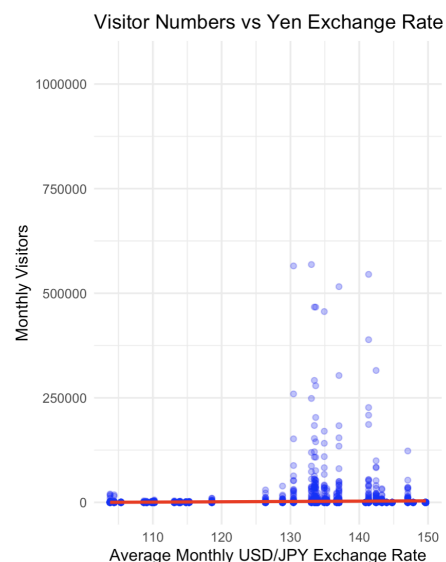
As for the Yen fluctuations, the dataset that was used showed daily Yen exchange rates. Therefore to make it simpler and match the COVID period format, the monthly average was calculated and formatted.

```
Call:
lm(formula = Visitor ~ AvgRate, data = merged)

Residuals:
    Min       1Q   Median       3Q      Max
-3720  -2739  -1366   -650  566159

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -7589.13   1899.85  -3.995 6.54e-05 ***
AvgRate       75.55     14.97   5.049 4.55e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 20140 on 7712 degrees of freedom
(9338 observations deleted due to missingness)
Multiple R-squared:  0.003294, Adjusted R-squared:  0.003165
F-statistic: 25.49 on 1 and 7712 DF, p-value: 4.553e-07
```



After running the model, the results show that the exchange rate coefficient is 75.55, meaning every 1 Yen value that decreases, around 75 people visit Japan more. Although the p-value is far below 0.05, and may seem statistically significant, the important aspect is the R-squared value of just 0.003, meaning that Yen rates alone does not explain even 1 percent of the reasoning of tourism. The residual standard error of 20140 confirms that it is not anywhere near significant or meaningful to conclude that Yen rates impact the number of visitors.

Conclusion

Although these simple models demonstrated how different variables impact tourism in Japan, there were clear limitations. One, the dependent variable itself isn't actually that clear, as the number of visitors doesn't necessarily equate to tourism. Reasons for visiting Japan may vary, therefore concluding that all of the visitors are there for tourism wouldn't be true. A better tailored research question could be considered. Second, the monthly visitors during Mid Covid was close to zero, making it difficult to normalize assumptions. Plots would show extremes, and were difficult to interpret because it was so abnormally low. Third, the model selection could

have been better. Linear regression alone for this project might not have been enough to clearly demonstrate the outcome, and perhaps a time series analysis may have been more appropriate.

To conclude, this simple project gave us an insight on how COVID periods and Yen exchange rates affected the number of visitors in Japan. It was clear from the results that the numbers have dropped significantly mid pandemic, and slowly started recovering post 2022. Yen exchange rates weren't as significant of a factor in our models, as the multiple R-squared value only showed around 0.003. Because this project had numerous limitations, a different model and a more detailed sample size, perhaps by including the daily numbers, could have been more appropriate, in order to achieve better results. We learned that the number of visitors were more based on policy changes, rather than economic shifts. Therefore we can reject the null hypothesis and accept the alternative (H1) hypothesis.

References

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